

DENSITY FUNCTIONS FOR THE EPSILON MULTIPLICITY OF HOMOGENEOUS IDEALS

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1. MULTIPLICITIES AND INTEGRAL CLOSURES

Let $(R, \mathfrak{m})$ be a Noetherian local ring of Krull dimension d .

1.1. Hilbert-Samuel multiplicity. Let $I \subseteq R$ be an $\mathfrak{m}$ -primary ideal. Then for all sufficiently large integers n , we have

$$\lambda_R(R/I^n) = P_I(n),$$

where $P_I(x)$ is a polynomial in x with rational coefficients of degree d . The *Hilbert-Samuel multiplicity* of I is defined by

$$e(I) = \lim_{n \rightarrow \infty} \frac{\lambda_R(R/I^n)}{n^d/d!} \in \mathbb{N}.$$

Equivalently, $e(I)$ is the normalized leading coefficient of the polynomial $P_I(x)$.

1.2. Integral closure. The *integral closure* $\bar{I}$ of an ideal I in R is defined as

$$\bar{I} = \{r \in R \mid r^n + a_1 r^{n-1} + \cdots + a_{n-1} r + a_n = 0 \text{ for some } n > 0, a_i \in I^i\}.$$

One can verify that $\bar{I}$ is again an ideal of R .

Theorem 1 (Rees 1960). *Let $(R, \mathfrak{m})$ be formally equidimensional and $I \subseteq J$ be two $\mathfrak{m}$ -primary ideals in R . Then*

$$\bar{J} = \bar{I} \text{ if and only if } e(I) = e(J).$$

1.3. j -multiplicity. Let $I \subseteq R$ be any ideal. The *j -multiplicity* of I is defined by

$$j(I) = \lim_{n \rightarrow \infty} \frac{\lambda_R(H_{\mathfrak{m}}^0(I^{n-1}/I^n))}{n^{d-1}/(d-1)!} \in \mathbb{N}.$$

This definition is due to Achilles and Manaresi.

1.4. ε -multiplicity. Let $I \subseteq R$ be any ideal. The *ε -multiplicity* of I is defined by

$$\varepsilon(I) = \limsup_{n \rightarrow \infty} \frac{\lambda_R(H_{\mathfrak{m}}^0(R/I^n))}{n^d/d!} = \limsup_{n \rightarrow \infty} \frac{\lambda_R((I^n :_R \mathfrak{m}^\infty)/I^n)}{n^d/d!}.$$

This definition is due to Ulrich and Validashti. We always have $0 \leq \varepsilon(I) \leq j(I)$.

Theorem 2 (Cutkosky 2014). *Let $(R, \mathfrak{m})$ be analytically unramified and $I \subseteq R$ be an ideal. Then $\varepsilon(I)$ exists as a limit.*

In general, the associated length function $n \mapsto \lambda_R(H_{\mathfrak{m}}^0(R/I^n))$ may not exhibit any *polynomial-like behaviour*.

Example 3 (Cutkosky, Ha, Srinivasan and Theodorescu 2005). *There exists a (homogeneous) prime ideal I in the polynomial ring $R = \mathbb{C}[x, y, z, w]$ such that $\varepsilon(I) \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$.*

The following theorem is a generalization of Rees' criterion to arbitrary ideals.

Theorem 4. *Let $(R, \mathfrak{m})$ be formally equidimensional and $I \subseteq J$ be two ideals in R . The following are equivalent.*

- (i) $\bar{I} = \bar{J}$.
- (ii) $\varepsilon(I_{\mathfrak{p}}) = \varepsilon(J_{\mathfrak{p}})$ for all $\mathfrak{p} \in \text{Ass}(R/\bar{J})$.
- (iii) $j(I_{\mathfrak{p}}) = j(J_{\mathfrak{p}})$ for all $\mathfrak{p} \in \text{Ass}(R/\bar{J})$.

Here, (i) $\iff$ (ii) is due to Ulrich and Validashti 2011, and (i) $\iff$ (iii) is due to Flenner and Manaresi 2001.

In the graded setup, we give a simple generalization of Rees' theorem that avoids localization.

Theorem 5 (–, Roy and Trivedi 2025). *Let $R = \bigoplus_{i \geq 0} R_i$ be a standard graded, finitely generated domain over a field $R_0 = k$. Let $I \subseteq J$ be two homogeneous ideals in R which are generated in equal degrees $d(I)$ and $d(J)$, respectively. Further, assume that the ideals have maximal analytic spreads. The following statements are equivalent.*

- (i) $\bar{I} = \bar{J}$.
- (ii) $d(I) = d(J)$ and $\varepsilon(I) = \varepsilon(J)$.
- (iii) $d(I) = d(J)$ and $j(I) = j(J)$.

2. DENSITY FUNCTIONS

Henceforth, we shall restrict ourselves to the following graded setup.

Notations 6. *Let k be an algebraically closed field, and let $R = \bigoplus_{m \geq 0} R_m$ be a standard graded, finitely generated domain over the field $R_0 = k$, with $\dim R = d \geq 2$. Let $\mathfrak{m}$ be the unique homogeneous maximal ideal of R , and let $I \subseteq R$ be a homogeneous ideal.*

Define the functions

$$f_I^{\text{sat}}(x) = \limsup_{n \rightarrow \infty} \frac{\dim_k (I^n :_R \mathfrak{m}^\infty)_{[xn]}}{n^{d-1}/(d-1)!}, \quad \text{and} \quad f_I(x) = \limsup_{n \rightarrow \infty} \frac{\dim_k (I^n)_{[xn]}}{n^{d-1}/(d-1)!},$$

where $x \in \mathbb{R}_{\geq 0}$. The functions f_I^{sat} and f_I are called the *saturation density function* and the *adic density function* of I , respectively.

2.1. Saturation density function. Consider the projective variety $V = \text{Proj } R$ with a very ample invertible sheaf $\mathcal{O}_V(1)$. Let $\mathcal{I}$ be the ideal sheaf associated to I on V . Let

$$\pi : X = \mathbf{Proj}(\bigoplus_{n \geq 0} \mathcal{I}^n) \longrightarrow V$$

be the blow up of V along $\mathcal{I}$. Then $\mathcal{I}$ becomes locally principal on X , i.e., there is an effective Cartier divisor E on X such that $\mathcal{I}\mathcal{O}_X = \mathcal{O}_X(-E)$. Let H be the pullback of a hyperplane section on V to X , so that $\mathcal{O}_X(H) = \pi^*\mathcal{O}_V(1)$.

Now define the constants

$$\begin{aligned} \alpha(I) &= \lim_{n \rightarrow \infty} \frac{\min\{m \in \mathbb{N} \mid (I^n :_R \mathfrak{m}^\infty)_m \neq 0\}}{n}, \\ s(I) &= \lim_{n \rightarrow \infty} \frac{\text{reg}(I^n :_R \mathfrak{m}^\infty)}{n}, \\ \rho(I) &= \lim_{n \rightarrow \infty} \frac{\text{reg}(I^n)}{n}. \end{aligned}$$

There exist inequalities

$$0 < \alpha(I) \leq s(I) \leq \rho(I).$$

Theorem 7 (–, Roy and Trivedi 2025). *Adopt Notations 6. The following statements hold.*

(1) For all $x \geq 0$,

$$f_I^{\text{sat}}(x) = \lim_{n \rightarrow \infty} \frac{\dim_k(I^n : R\mathfrak{m}^\infty)_{\lfloor xn \rfloor}}{n^{d-1}/(d-1)!} = \text{vol}_X(xH - E),$$

where $\text{vol}_X : N^1(X)_{\mathbb{R}} \rightarrow \mathbb{R}$ is the volume function in the sense of Lazarsfeld.

(2) $f_I^{\text{sat}}(x) = 0$ for all $x \leq \alpha(I)$.

(3) For all $x \geq s(I)$,

$$f_I^{\text{sat}}(x) = (xH - E)^{d-1} = \sum_{i=0}^{d-1} (-1)^{d-1-i} \binom{d-1}{i} (H^i \cdot E^{d-1-i}) x^i.$$

(4) The function f_I^{sat} is continuous on $\mathbb{R}_{\geq 0}$, and is continuously differentiable on $\mathbb{R}_{\geq 0} \setminus \{\alpha(I)\}$.

(5) On the interval $(\alpha(I), \infty)$, f_I^{sat} is positive, strictly increasing, and log-concave, i.e.,

$$(f_I^{\text{sat}}(x+y))^{\frac{1}{d-1}} \geq (f_I^{\text{sat}}(x))^{\frac{1}{d-1}} + (f_I^{\text{sat}}(y))^{\frac{1}{d-1}} \quad \forall x, y \geq \alpha(I).$$

(6) For all integers $c > 0$,

$$d \cdot \int_0^c f_I^{\text{sat}}(x) dx = \lim_{n \rightarrow \infty} \frac{\sum_{m=0}^{cn} \dim_k(I^n : R\mathfrak{m}^\infty)_m}{n^d/d!}.$$

2.2. Adic density function.

Theorem 8 (–, Roy and Trivedi 2025). *Adopt Notations 6. Let d_1 denote the least generating degree of I . The following statements hold.*

(1) For all $x \geq 0$, $f_I(x)$ exists as a limit.

(2) There exist finitely many integers

$$0 < d_1 < d_2 < \cdots < d_r := \rho(I)$$

such that

$$f_I(x) = \begin{cases} 0 & \text{if } x \in [0, d_1), \\ P_1(x) & \text{if } x \in (d_1, d_2), \\ \vdots & \vdots \\ P_{r-1}(x) & \text{if } x \in (d_{r-1}, d_r), \\ P_r(x) & \text{if } x \in (d_r, \infty), \end{cases}$$

where $P_i(x)$ is a polynomial in x with rational coefficients and $\deg P_i(x) \leq d-1$.

Moreover, f_I is continuous on $\mathbb{R}_{\geq 0} \setminus \{d_1\}$, and $P_r(x) = (xH - E)^{d-1}$.

(3) On the interval (d_1, ∞) , f_I is positive, strictly increasing, and log-concave, i.e.,

$$(f_I(x+y))^{\frac{1}{d-1}} \geq (f_I(x))^{\frac{1}{d-1}} + (f_I(y))^{\frac{1}{d-1}} \quad \forall x, y > d_1.$$

(4) For all integers $c > 0$,

$$d \cdot \int_0^c f_I(x) dx = \lim_{n \rightarrow \infty} \frac{\sum_{m=0}^{cn} \dim_k(I^n)_m}{n^d/d!}.$$

2.3. Epsilon density function.

Theorem 9 (–, Roy and Trivedi 2025). *Adopt Notations 6. The following statements hold.*

(1) *For all $x \geq 0$,*

$$f_I^\varepsilon(x) := \lim_{n \rightarrow \infty} \frac{\dim_k (H_m^0(R/I^n))_{\lfloor xn \rfloor}}{n^{d-1}/(d-1)!} = f_I^{\text{sat}}(x) - f_I(x).$$

(2) *The function f_I^ε is zero outside the compact interval $[\alpha(I), \rho(I)]$.*

(3) *The function f_I^ε is continuous on $\mathbb{R}_{\geq 0} \setminus \{\alpha(I)\}$. Moreover, it is continuously differentiable except for finitely many values of x .*

(4) *We have*

$$d \cdot \int_0^\infty f_I^\varepsilon(x) dx = \lim_{n \rightarrow \infty} \frac{\lambda_R(H_m^0(R/I^n))}{n^d/d!} = \varepsilon(I).$$

Corollary 10 (–, Roy and Trivedi 2025). *Let R be as in Notations 6. Let $I \subseteq R$ be a homogeneous ideal such that I is generated in equal degrees $d(I)$ and $1 \leq \text{height } I \leq d-1$. Then*

$$f_I(x) = \begin{cases} 0 & \text{if } x < d(I), \\ (xH - E)^{d-1} & \text{if } x \geq d(I), \end{cases}$$

and

$$\varepsilon(I) = d \cdot \int_0^{d(I)} \text{vol}_X(xH - E) dx.$$

2.4. Density functions and integral closures.

Theorem 11 (–, Roy and Trivedi 2025). *Let R be as in Notations 6, and $I \subseteq J$ be two homogeneous ideals in R . The following statements are equivalent.*

- (i) $\bar{I} :_R \mathbf{m}^\infty = \bar{J} :_R \mathbf{m}^\infty$.
- (ii) $f_I^{\text{sat}}(x) = f_J^{\text{sat}}(x)$ for all $x \in \mathbb{R}_{\geq 0}$.
- (iii) $f_I^{\text{sat}}(x) = f_J^{\text{sat}}(x)$ for some $x > \max\{s(I), s(J)\}$.
- (iv) $f_I(x) = f_J(x)$ for some $x > \max\{\rho(I), \rho(J)\}$.

Theorem 12 (–, Roy and Trivedi 2025). *Let R be as in Notations 6, and $I \subseteq J$ be two homogeneous ideals in R . The following statements are equivalent.*

- (i) $\bar{I} = \bar{J}$.
- (ii) $f_I(x) = f_J(x)$ for all $x \in \mathbb{R}_{\geq 0}$.
- (iii) $\varepsilon(I) = \varepsilon(J)$ and $f_I(x) = f_J(x)$ for some $x > \max\{\rho(I), \rho(J)\}$.
- (iv) $j(I) = j(J)$ and $f_I(x) = f_J(x)$ for some $x > \max\{\rho(I), \rho(J)\}$.

2.5. Examples of saturation density functions.

Theorem 13 (–, Dubey, Roy and Verma 2025). *Let R be as in Notations 6 with $\dim R = 2$. Let $I \subseteq R$ be a height one homogeneous ideal. Then*

$$f_I^{\text{sat}}(x) = \begin{cases} 0 & \text{if } x \leq \frac{a}{b}, \\ bx - a & \text{if } x \geq \frac{a}{b}. \end{cases}$$

where $\alpha(I) = s(I) = \frac{a}{b}$ is a positive rational number.

Theorem 14 (–, Roy and Trivedi 2025). *Let R be as in Notations 6 with $\dim R = 3$. Let $I \subseteq R$ be a homogeneous ideal such that $1 \leq \text{height } I \leq 2$. Then there exists a real quadratic polynomial $Q(x)$ such that*

$$f_I^{\text{sat}}(x) = 0 \quad \forall x \in [0, \alpha(I)], \quad \text{and} \quad f_I^{\text{sat}}(x) = Q(x) \quad \forall x \in [s(I), \infty).$$

Moreover, if $\alpha(I) < s(I)$, then

$$f_I^{\text{sat}}: (\alpha(I), s(I)] \longrightarrow \mathbb{R}_{\geq 0}$$

is locally given by a rational polynomial function of degree at most two.

Example 15 (Nagata). *Let $R = \mathbb{C}[x, y, z]$ be the standard graded polynomial ring in the variables x, y, z over $R_0 = \mathbb{C}$. Let $s \geq 16$ be an integer such that s is a perfect square, and consider s independent generic points $p_1, \dots, p_s$ in $\mathbb{P}_{\mathbb{C}}^2$. Let I be the ideal generated by all homogeneous polynomials in R that vanish simultaneously at $p_1, \dots, p_r$. Then*

$$f_I^{\text{sat}}(x) = \begin{cases} 0 & \text{if } x \leq \sqrt{s}, \\ x^2 - s & \text{if } x \geq \sqrt{s}. \end{cases}$$

Example 16 (– and Sengupta). *There exists a 4-dimensional standard graded Noetherian domain $R = \bigoplus_{m \geq 0} R_m$ over $R_0 = \mathbb{C}$ and a height one homogeneous ideal $I \subseteq R$ such that*

- (1) R is normal, generalized Cohen-Macaulay and $\text{Proj } R$ is nonsingular,
- (2) $\text{Proj } R/I$ is nonsingular (which implies that I is radical), and
- (3) the saturation density function

$$f_I^{\text{sat}}(x) = \begin{cases} 0 & \text{if } x \leq \frac{3(9+\sqrt{5})}{38}, \\ P(x) + Q(x)\sqrt{17x^2 - 36x + 36} & \text{if } \frac{3(9+\sqrt{5})}{38} \leq x \leq 3, \\ R(x) & \text{if } x \geq 3. \end{cases}$$

where $P(x)$, $Q(x)$ and $R(x)$ are polynomials in $\mathbb{Q}[x]$ of degrees 3, 2 and 3 respectively.

In particular, $\varepsilon(I) \in \mathbb{R} \setminus \overline{\mathbb{Q}}$.